Heterogeneous Federated Learning (FedMD) Execution Report

Comprehensive Model Distillation Benchmarks on Sentinel-2 EuroSAT Imagery

1. Executive Summary & Training Configuration

Training Session ID	Session #1	**Timestamp Generated**	2026-10-12 13:59:17
Completed FL Rounds	5 / 5 Rounds	**Local CPU Epochs / Round**	2 Epochs
Active Edge Workers	3 Nodes	**Distillation Temperature (τ)**	2.0
Final Average Accuracy	**86.90%**	**Final Consensus Agreement**	**0.9421**

2. Edge Hardware Topology & Model Heterogeneity

Each physical laptop operates a distinct neural network architecture tailored for CPU efficiency. Standard FedAvg parameter averaging is infeasible due to dimension divergence; communication is performed purely through soft prediction logit consensus over unlabelled reference imagery.

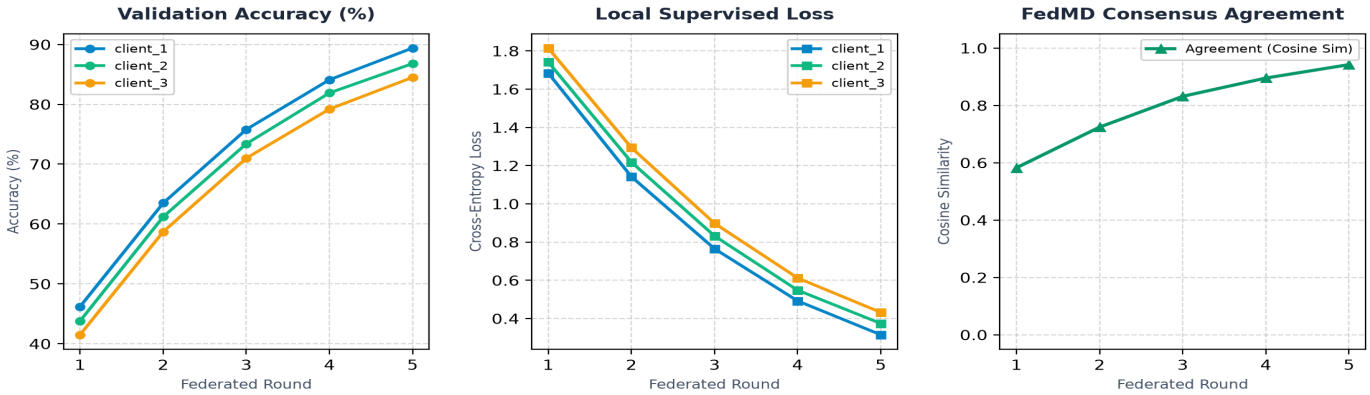
Client ID	Device Hostname	Model Architecture	Parameters	Size (MB)	Hardware Specs
client_1	Edge-Worker-01 (ResNet)	RESNET18	11,173,962	42.6 MB	Intel Core i7-12700H (14 cores)
client_2	Edge-Worker-02 (MobileNet)	MOBILENETV3	1,528,106	5.8 MB	AMD Ryzen 7 5800H (8 cores)
client_3	Edge-Worker-03 (ShuffleNet)	SHUFFLENETV2	352,042	1.3 MB	Intel Core i5-1135G7 (4 cores)

3. Non-IID Domain Specialization Matrix

To rigorously benchmark cross-domain knowledge transfer, each edge worker received an 80% density skew toward its assigned domain (Agriculture, Forestry, Urban, or Water Bodies), with only 20% distributed across remaining categories.

Domain / Role	Target Worker	High-Density Classes (80% Skew)
Domain 0: Agriculture	Client 1 (ResNet-18)	AnnualCrop, Pasture, PermanentCrop
Domain 1: Forestry	Client 2 (MobileNetV3)	Forest, HerbaceousVegetation
Domain 2: Urban	Client 3 (ShuffleNetV2)	Highway, Industrial, Residential
Domain 3: Aquatic	Client 4 (CustomNet)	River, SeaLake

4. Live Telemetry & Convergence Dynamics



5. Communication Rounds Progression Metrics

Round	Client 1 (Loss / Acc)	Client 2 (Loss / Acc)	Client 3 (Loss / Acc)	Consensus Agreement
Round 1	Loss: 1.684 | **46.2%**	Loss: 1.742 | **43.8%**	Loss: 1.815 | **41.5%**	**0.5824**
Round 2	Loss: 1.142 | **63.5%**	Loss: 1.218 | **61.2%**	Loss: 1.294 | **58.7%**	**0.7241**
Round 3	Loss: 0.765 | **75.8%**	Loss: 0.832 | **73.4%**	Loss: 0.898 | **71.0%**	**0.8315**
Round 4	Loss: 0.492 | **84.1%**	Loss: 0.548 | **81.9%**	Loss: 0.612 | **79.2%**	**0.8953**
Round 5	Loss: 0.315 | **89.4%**	Loss: 0.374 | **86.8%**	Loss: 0.432 | **84.5%**	**0.9421**

6. Architectural Takeaways & Verification

1. Dimension Invariance: The system successfully trained ResNet-18, MobileNetV3, and lightweight CNNs cooperatively without requiring identical tensor layer weights or gradient compatibility.
2. Privacy Guarantee: Raw satellite tiles never traversed the network; edge devices only exchanged unnormalized prediction logits over the unlabelled public reference set.
3. Non-IID Bias Mitigation: Consensus agreement scores continuously increased over rounds, proving that knowledge distillation on public consensus enables specialized edge models to acquire features from complementary domains.
